

Three Roads to the Cosmological-Constant Acceleration Scale: Dynamics, Weak-Lensing Geometry, and Expansion — and Why the Lensing Leg Is a Consistency, Not Yet a Free-Fit

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Abstract

A single low-acceleration scale $a_0 \sim 1 \times 10^{-10} \text{ m s}^{-2}$ organizes galaxy rotation curves (the mass-discrepancy–acceleration / radial-acceleration relation), and its numerical coincidence with $c\sqrt{\Lambda}$ has been noted since Milgrom’s original papers. In the de Sitter–Unruh modified-inertia framework used here, that coincidence is promoted to an identity, $a_0 = cH_\Lambda/Z$ with $H_\Lambda = c\sqrt{\Lambda/3}$ and $Z = \sqrt{32\pi/3} = 5.78881$ *posited*, which inverts to $\Lambda = 32\pi a_0^2/c^4$. This note asks whether *three* observational channels with mutually **orthogonal systematics** independently land on the same scale and hence on the same Λ : (i) **dynamics** — the SPARC rotation-curve a_0 (kinematic systematics: stellar M/L , inclination, asymmetric drift); (ii) **geometry** — the KiDS-1000 weak-lensing radial-acceleration relation of Brouwer et al. (2021), which reaches $g_{\text{bar}} \sim 10^{-15} \text{ m s}^{-2}$, about five decades below a_0 and three decades deeper than any rotation curve (shear-calibration and photo- z systematics); and (iii) **expansion** — the a_0 demanded by the Type Ia supernova Hubble diagram through the Friedmann dark-energy term (standardization systematics). The three central values span $\approx (0.96\text{--}1.20) \times 10^{-10} \text{ m s}^{-2}$ (a \$22% spread; \$30% including the low edge of the SNe range, 0.92×10^{-10}), all inside the SPARC systematics box $[0.84, 1.36] \times 10^{-10}$, and all invert to Λ within a factor \$1.6 of the Planck value (KiDS $a_0 = 1.20 \times 10^{-10} \Rightarrow \Lambda = 1.79 \times 10^{-52} \text{ m}^{-2} = 1.65 \times \Lambda_{\text{Planck}}$; Planck sits 0.91σ from the lensing central value). **We state plainly that this is a consistency + method + forecast note, not a free-fit measurement.** The KiDS leg is a *consistency*: Brouwer adopted a_0 (from McGaugh 2016) and tested agreement; a genuine lensing-only free-fit is not yet possible from public data and, from the figure alone, is reading- and scatter-limited to a factor \$2. A tabulated free-fit does not shortcut this: because $a_0 \propto g_{\text{obs}}^2$ amplifies the per-point scatter, the statistical error alone is \$13–43 \$43% for the clean reliable sample), comparable to the \$15–25% systematic floor (the $\geq 6\sigma$ early/late-type split; the baryonic-mass/deprojection model), so a realistic clean-sample free-fit reaches only \$45a_0 = \$ dynamical- a_0 is a *prediction of that completion*, not automatic. The value of a_0 and Z are posited, not derived.

1. Introduction

The radial-acceleration relation (RAR) states that the observed centripetal acceleration in a galaxy, g_{obs} , is a tight one-to-one function of the acceleration g_{bar} implied by the baryons alone (Milgrom 1983; McGaugh, Lelli & Schombert 2016; Lelli, McGaugh & Schombert 2016). The relation carries a single scale $a_0 \simeq 1.2 \times 10^{-10} \text{ m s}^{-2}$ separating the Newtonian regime $g_{\text{bar}} \gg a_0$ from the low-acceleration regime $g_{\text{bar}} \ll a_0$ where $g_{\text{obs}} \rightarrow \sqrt{a_0 g_{\text{bar}}}$. It has been noticed since Milgrom’s earliest work that a_0 is numerically of order cH_0 and of order $c\sqrt{\Lambda}$, i.e. that the dynamical scale coincides with the cosmological-constant scale.

The framework used here takes that coincidence literally. It is a **modified-inertia** theory in which the de Sitter–Unruh temperature of the cosmic horizon sets an inertial floor, giving

$$a_0 = \frac{cH_\Lambda}{Z}, \quad H_\Lambda \equiv c\sqrt{\frac{\Lambda}{3}}, \quad Z = \sqrt{\frac{32\pi}{3}} = 5.78881 \text{ (posited)},$$

with the low-acceleration kernel

$$g_{\text{obs}} = \sqrt{g_{\text{bar}}^2 + g_{\text{bar}} a_0},$$

which is algebraically Milgrom’s (1999, PLA **253**, 273) ν -kernel $\nu(y) = \sqrt{1 + 1/y}$; the framework’s distinctive content is the coefficient cH_Λ/Z (Milgrom’s coincidence used $\sim 2cH_0$) together with the modified-inertia completion. Eliminating H_Λ gives the inversion used throughout this note,

$$\Lambda = \frac{32\pi a_0^2}{c^4}$$

Because Λ scales as a_0^2 , a factor- f uncertainty on a_0 becomes a factor- f^2 uncertainty on Λ ; conversely a modest \$30% spread in a_0 is a factor \$1.6 in Λ . We stress at the outset that neither Z nor the *value* of a_0 is derived here — both are posited — so what follows tests the *universality of the scale and its inversion to Λ* , not the origin of the number.

The idea of this paper is simple. If a_0 is a genuine constant of nature tied to Λ , then observables whose systematic errors are unrelated should agree on it. We assemble **three channels**:

- **Dynamics** — rotation-curve kinematics (SPARC).
- **Geometry** — weak gravitational lensing (KiDS-1000).
- **Expansion** — the supernova Hubble diagram.

These three have essentially orthogonal error budgets. Agreement among them is a *cross-check that Λ CDM does not automatically supply*, because in Λ CDM

the galactic a_0 is an emergent feature of feedback-regulated halo assembly with no a-priori link to the value of Λ in the Friedmann equation. **This is a consistency + method + forecast note.** We show the triangle closes at the $\sim 30\%$ / factor-1.6 level today, we explain honestly why the lensing leg is presently a *consistency* rather than a free-fit, and we forecast what a proper tabulated free-fit would deliver. Every dimensional number below is reproduced by the committed script `kids_rar_lambda.py` (exit 0) and is quoted for both a_0 footings.

Two footings for a_0 are carried throughout, as the framework requires: the **canonical** footing $a_0 = cH_\Lambda/Z = 9.355 \times 10^{-11} \text{ m s}^{-2}$ (in which Λ equals the Planck value by construction), and the **alternative** footing $a_0 = 1.1305 \times 10^{-10}$ (built from ρ_{total} and cH_0 rather than ρ_{DE} and cH_Λ). The measured channels are quoted as measured and then inverted; the footings bracket where the theory *expects* a_0 to sit.

2. The three channels and their orthogonal systematics

2.1 Dynamics — the SPARC rotation-curve a_0

The sharpest single-number extraction of a_0 from rotation curves in this program is the “ a_0 -line” estimator (Zimmerman, DOI 10.5281/zenodo.21419735), which uses the identity

$$g_{\text{obs}}^2 - g_{\text{bar}}^2 = a_0 g_{\text{bar}}$$

(exact for the framework kernel) and reads a_0 off the gas-dominated slope, where the stellar M/L , distance, and inclination dependences largely cancel in the pairwise estimator. The result is

$$a_0^{\text{dyn}} = 1.181 \times 10^{-10} \text{ m s}^{-2}, \quad \text{systematics box } [0.84, 1.36] \times 10^{-10}.$$

The box is dominated by *kinematic* systematics: the stellar mass-to-light ratio Υ_* , disk inclination, and asymmetric-drift / pressure-support corrections. This is the dynamics leg.

2.2 Geometry — the KiDS-1000 weak-lensing RAR

Brouwer et al. (2021, A&A **650**, A113; arXiv:2106.11677) constructed the RAR from *weak gravitational lensing* around isolated galaxies in the KiDS-1000 survey, cross-matched with GAMA and with photometric (“KiDS-bright”) lens samples. Excess-surface-density profiles convert to $g_{\text{obs}}(R)$ out to $\sim \text{Mpc}$ scales, so the lensing RAR spans

$$10^{-15} \lesssim g_{\text{bar}} \lesssim 5 \times 10^{-12} \text{ m s}^{-2},$$

roughly **five decades below** a_0 and about **three decades deeper than the deepest rotation curve**. In 15 logarithmic bins Brouwer show that the *reliable* (spectroscopic GAMA) lensing RAR agrees with the MOND/modified-gravity prediction at the adopted $g_{\dagger} = a_0 = (1.20 \pm 0.26) \times 10^{-10}$ (McGaugh 2016). The systematics — shear calibration, photometric-redshift errors, source blending, boost/contamination corrections — are *orthogonal* to the kinematic systematics of the dynamics leg. This is the geometry leg, and its numerical value used below is the Brouwer-adopted $a_0 = (1.20 \pm 0.26) \times 10^{-10}$.

2.3 Expansion — the SNe-demanded a_0

In the same framework the Friedmann dark-energy term carries a_0 through $\Omega_{\Lambda}^2 a_0^2/c^2 = \Omega_{\Lambda}$ (dark-energy density scale), so the Type Ia supernova Hubble diagram *demands* an a_0 once H_0 is fixed. From the cross-scale $a_0(z)$ analysis (Zimmerman, DOI 10.5281/zenodo.21440407; the `side_by_side` construction), the Pantheon+ diagram over the plausible range $H_0 = 67.4 \rightarrow 73 \text{ km s}^{-1} \text{ Mpc}^{-1}$ demands

$$a_0^{\text{exp}}(z=0) = (9.17\text{--}9.93) \times 10^{-11} \text{ m s}^{-2} = (0.92\text{--}0.99) \times 10^{-10}.$$

The dominant systematics here are supernova standardization, calibration of the distance ladder, and the choice of H_0 — again orthogonal to both other legs. This is the expansion leg.

2.4 The triangle and the Λ inversions

Collecting the three central values (in units of $10^{-10} \text{ m s}^{-2}$):

Channel	a_0	Systematics
Dynamics (SPARC a_0 -line)	1.18, box [0.84, 1.36]	Υ_* , inclination, asymmetric drift
Geometry (KiDS lensing)	1.20 ± 0.26	shear calibration, photo- z
Expansion (SNe-demanded)	0.92–0.99	SNe standardization, H_0
<i>framework canonical</i>	0.94	$\Lambda = \Lambda_{\text{Planck}}$ by construction

The three measured central values span $\approx (0.96\text{--}1.20) \times 10^{-10}$ (a $\approx 22\%$ spread on centrals); including the low edge of the SNe range the full span is $\approx (0.92\text{--}1.20) \times 10^{-10}$, i.e. $\approx 22\text{--}30\%$ wide. **All three lie inside the SPARC**

systematics box $[0.84, 1.36]$, and all invert to Λ within a factor ~ 1.6 of the Planck value.

Working the geometry leg explicitly through $\Lambda = 32\pi a_0^2/c^4$:

$$a_0^{\text{geo}} = 1.20 \times 10^{-10} \Rightarrow \Lambda_{\text{lens}} = 1.792 \times 10^{-52} \text{ m}^{-2}, \quad \text{range } [1.100, 2.653] \times 10^{-52},$$

against $\Lambda_{\text{Planck}} = 1.089 \times 10^{-52} \text{ m}^{-2}$. The ratio is $\Lambda_{\text{lens}}/\Lambda_{\text{Planck}} = 1.65$ (range $[1.01, 2.44]$), and the Planck value sits 0.91σ from the lensing central value — **consistent**, and on the same “high side of the box” as the SPARC dynamics leg. (This 0.91σ is computed in Λ -space with a linearized half-width; the equivalent comparison in a_0 -space — Planck-equivalent $a_0 = 9.354 \times 10^{-11}$ against 1.20 ± 0.26 — is 1.02σ . Both are well within 2σ ; we quote both frames so the Λ -space number is not read as the whole story.) In the canonical footing $a_0 = 9.355 \times 10^{-11}$ reproduces Λ_{Planck} exactly by construction; in the alternative footing $a_0 = 1.1305 \times 10^{-10}$ gives $\Lambda = 1.59 \times 10^{-52} = 1.46 \times \Lambda_{\text{Planck}}$. Three observables with orthogonal systematics agreeing on a_0 to $\sim 30\%$ is not a precision match — but it is a genuinely independent third leg on the statement “the scale is universal at $\sim \sqrt{\Lambda}$.”

3. The lensing leg in depth

3.1 What Brouwer measured, and the five-decade reach

The lensing RAR is qualitatively different from the kinematic RAR in *where it lives*. Rotation curves probe g_{bar} from $\sim \text{few} \times 10^{-9}$ down to at best $\sim 10^{-12} \text{ m s}^{-2}$ in the most extended H, {1} DISKS — I.E. BARELY ONE TO TWO DECADES BELOW a_0 . THE KIDS LENSING PROFILES, BY STACKING EXCESS SURFACE DENSITY OUT TO MPC RADII, PUSH g_{bar} DOWN TO $\sim 10^{-15} \text{ m s}^{-2}$, ROUGHLY FIVE DECADES BELOW a_0 . THIS MATTERS FOR THE ACCELERATION-SCALE PROGRAM BECAUSE THE DEEP REGIME IS WHERE THE INTERPOLATION-FUNCTION AMBIGUITY DISAPPEARS.

3.2 Interpolation-freeness of the deep-regime extraction

EVERY LENSING BIN SITS DEEP IN THE LOW-ACCELERATION REGIME ($g_{\text{bar}} \ll a_0$), WHERE ALL VIABLE INTERPOLATION KERNELS COLLAPSE ONTO THE SAME ASYMPTOTE $g_{\text{obs}} \rightarrow \sqrt{a_0 g_{\text{bar}}}$, SO $a_0 = g_{\text{obs}}^2/g_{\text{bar}}$ INDEPENDENT OF KERNEL CHOICE. EVALUATED AT $g_{\text{bar}} = 10^{-15} \text{ m s}^{-2}$ ($y \equiv g_{\text{bar}}/a_0 = 8.3 \times 10^{-6}$):

KERNEL	$g_{\text{obs}} \text{ (M s}^{-2}\text{)}$
FRAMEWORK ν -KERNEL	3.464116×10^{-13}
$\sqrt{g_{\text{bar}}^2 + g_{\text{bar}} a_0}$	
BROUWER “THIRD-FAMILY”	3.469104×10^{-13}
$g_{\text{bar}}/(1 - e^{-\sqrt{y}})$	
PURE DEEP-MOND $\sqrt{a_0 g_{\text{bar}}}$	3.464102×10^{-13}

THE MAXIMUM FRACTIONAL SPREAD ACROSS KERNELS IS 1.44×10^{-3} — ABOUT **0.14%**. FIVE DECADES DEEP, THE EXTRACTION $a_0 = g_{\text{obs}}^2/g_{\text{bar}}$ IS INTERPOLATION-FREE TO WELL BELOW ANY OBSERVATIONAL ERROR, AND IT RECOVERS 1.200×10^{-10} EXACTLY. THIS IS THE GEOMETRY CHANNEL’S STRUCTURAL EDGE OVER ROTATION CURVES, WHICH LIVE NEAR $y \sim 1$ WHERE THE KERNEL CHOICE IS A REAL (\$ \$10–20%) SYSTEMATIC. NO BARYONIC EFFECT CAN MIMIC a_0 THIS FAR INTO THE DEEP REGIME EITHER — THE BARYONS ARE $\sim$ FIVE DECADES SUBDOMINANT.

3.3 Why a figure free-fit is reading- and scatter-limited (honest scope)

IT IS TEMPTING TO DIGITIZE BROUWER’S FIG. 4 AND FREE-FIT a_0 OURSELVES. WE DO **not** PRESENT SUCH A NUMBER AS A MEASUREMENT, FOR CONCRETE REASONS. FIG. 4 IS DEEP-MOND THROUGHOUT, SO THE DATA LIE ALONG A SLOPE- $\frac{1}{2}$ LINE AND $a_0 = g_{\text{obs}}^2/g_{\text{bar}}$; A SMALL VERTICAL READING ERROR PROPAGATES AS $\log a_0 = 2 \log g_{\text{obs}} - \log g_{\text{bar}}$, I.E. **doubled**. TWO FURTHER PROBLEMS COMPOUND THIS:

- **The KiDS-bright (photometric) sample is selection-biased high.** THAT THE PHOTOMETRIC KiDS-BRIGHT LENS SAMPLE CARRIES A PHOTO- z SELECTION BIAS IS BROUWER’S OWN CAVEAT (THEIR §-LEVEL DISCUSSION OF THE BRIGHT-SAMPLE SELECTION, ARXIV:2106.11677); THE SPECIFIC NUMBER IS *OUR* READING, NOT THEIRS. BY-EYE DIGITIZATION OF THAT SAMPLE GIVES $a_0 \sim 1\text{--}3 \times 10^{-10}$ (CENTERED NEAR $\sim 2 \times 10^{-10}$), WHICH WE ATTRIBUTE TO THAT SELECTION BIAS PLUS \$ \$0.1-DEX READING ERROR — AN ARTIFACT, NOT A MEASUREMENT.
- **The reliable spectroscopic GAMA sample has large intrinsic scatter.** IT TRACKS MOND AT $a_0 \sim 1.2 \times 10^{-10}$, BUT WITH \$ \$0.2–0.4 DEX PER POINT OF INTRINSIC SCATTER, SO EVEN WITH *PERFECT* READING A FIGURE FREE-FIT CONSTRAINS a_0 ONLY TO A **factor \$ \$2**.

THIS IS PRECISELY WHY BROUWER **fixed** a_0 (ADOPTING MCGAUGH’S VALUE) AND TESTED *CONSISTENCY* RATHER THAN FREE-FITTING IT. OUR KiDS LEG INHERITS THAT STATUS: IT IS A CONSISTENCY CHECK, NOT AN INDEPENDENT FREE-FIT, AND WE PRESENT IT AS SUCH.

4. The forecast: what a tabulated free-fit would deliver

A GENUINE LENSING-ONLY FREE-FIT NEEDS THE **per-point tabulated** $(g_{\text{bar}}, g_{\text{obs}}, \sigma)$ DATA, NOT THE FIGURE. THAT TABLE IS NOT CLEANLY PUBLIC: THERE IS NO VIZIER ENTRY, AND THE KiDS DATA PORTAL EXPOSES ONLY RAW EXCESS-SURFACE-DENSITY PROFILES THAT REQUIRE A FULL LENSING-INVERSION PIPELINE TO TURN INTO $(g_{\text{bar}}, g_{\text{obs}})$ POINTS. WITH THE TABLE IN HAND, CONSIDER THE STATISTICAL PRECISION FROM A SLOPE- $\frac{1}{2}$ INTERCEPT OVER $N = 15$ BINS. IN THE DEEP REGIME $\log g_{\text{obs}} = \frac{1}{2}(\log a_0 + \log g_{\text{bar}})$, SO THE INTERCEPT HAS SENSITIVITY $\partial \log g_{\text{obs}} / \partial \log a_0 = \frac{1}{2}$ AND $a_0 = g_{\text{obs}}^2 / g_{\text{bar}}$ **amplifies** THE PER-POINT ERROR BY A FACTOR 2:

$$\sigma_{\log a_0}^{\text{stat}} \approx \frac{2 \sigma_{\text{point}}}{\sqrt{N}} \Rightarrow \begin{cases} \text{KiDS-BRIGHT, } \sigma_{\text{point}} = 0.10 \text{ DEX :} & 0.052 \text{ DEX} \approx 13\% \\ \text{GAMA RELIABLE, } \sigma_{\text{point}} = 0.30 \text{ DEX :} & 0.155 \text{ DEX} \approx 43\% \end{cases}$$

SO STATISTICAL ALONE IS \$13% BIASED-HIGH) BRIGHT SAMPLE AND \$43% FOR THE RELIABLE GAMA SAMPLE — THE FACTOR $a_0 \propto g_{\text{obs}}^2$ AMPLIFIES RATHER THAN SUPPRESSES THE SCATTER, AND THE REAL FITTER `FIT_KIDS_RAR.PY` ON SYNTHETIC \$0.08-DEX DATA CONFIRMS THIS ORDER (TENS OF PERCENT, NOT A FEW PERCENT). TWO SYSTEMATICS SIT ON TOP:

- THE $\geq 6\sigma$ EARLY/LATE-TYPE RAR SPLIT (SECTION 5) MEANS A SINGLE UNIVERSAL a_0 IS A MODEL CHOICE, NOT A DATUM;
- THE BARYONIC-MASS ESTIMATE AND THE DEPROJECTION / HALO-MODEL USED TO TURN STACKED $\Delta\Sigma$ INTO g_{obs} CARRY THEIR OWN MULTIPLICATIVE UNCERTAINTY.

TAKING THE SYSTEMATIC FLOOR AT \$15–25% AND ADDING IN QUADRATURE, STATISTICAL AND SYSTEMATIC ARE **comparable** — **neither dominates**. A REALISTIC TABULATED FREE-FIT REACHES a_0 TO \$20–28% FOR THE BIASED BRIGHT SAMPLE AND \$45–50 *is a modest for the clean case* (\$45%), NOT THE NAIVE \$20%. IT IS WORTH DOING WHEN THE PER-POINT TABLE BECOMES PUBLIC, BUT IT IS NOT A FAST ROUTE TO A PERCENT-LEVEL a_0 . **This paper books the method and the triangle now**; THE FREE-FIT IS DEFERRED TO DATA AVAILABILITY.

5. The $\geq 6\sigma$ early/late-type split

BROUWER FIND THAT EARLY- AND LATE-TYPE GALAXIES FOLLOW RARS THAT DIFFER BY $\geq 6\sigma$ AT FIXED STELLAR MASS M_* : AT A GIVEN M_* THE EARLY-TYPES SHOW *EXCESS* g_{obs} . BROUWER’S OWN READING IS THAT EARLY-TYPES CARRY ADDITIONAL UNSEEN BARYONS — CIRCUMGALACTIC GAS WITH $M_{\text{gas}} \sim M_*$ — SO THAT THE BARYONIC g_{bar} IS UNDERESTIMATED FOR THEM AND THE

“EXCESS” IS A MISSING-BARYON ARTIFACT RATHER THAN A VIOLATION OF A UNIVERSAL a_0 .

EITHER WAY, THIS IS A **real systematic that must be carried, not foot-noted**. IF A SINGLE a_0 IS FIT ACROSS BOTH POPULATIONS, THE SPLIT BIASES AND BROADENS THE INFERRED VALUE; IF THE EXCESS IS TRULY CIRCUMGALACTIC GAS, THEN THE *CLEAN* CASE FOR THE ACCELERATION-SCALE PROGRAM IS THE **late-type disk sample**, WHERE THE BARYON CENSUS IS MOST COMPLETE AND THE DEEP-REGIME LENSING POINTS ARE LEAST CONTAMINATED. OUR RECOMMENDED FREE-FIT TARGET (SECTION 4) IS THEREFORE THE LATE-TYPE / DISK-SELECTED RELIABLE SAMPLE, AND THE \$15–25% SYSTEMATIC FLOOR ALREADY REFLECTS THE COST OF THIS SPLIT. WE DO NOT CLAIM THE SPLIT IS RESOLVED; WE CLAIM IT IS BOUNDED AND THAT THE LATE-TYPE CASE IS THE DEFENSIBLE ONE.

6. The disformal wrinkle: the honest limit on independence

THERE IS A FRAMEWORK-SPECIFIC SUBTLETY THAT SETS THE *HONEST CEILING* ON HOW “INDEPENDENT” THE GEOMETRY LEG CAN BE. IN ORDINARY SINGLE-METRIC MODIFIED GRAVITY (E.G. AN AeST/RELATIVISTIC-MOND REALIZATION), PHOTONS AND MATTER SHARE ONE METRIC, SO A LENSING RAR AND A DYNAMICAL RAR TEST THE SAME FUNCTION AND LENSING- $a_0 = \text{DYNAMICAL-}a_0$ IS ESSENTIALLY TRIVIAL. IN **modified inertia**, THIS IS NOT THE CASE.

HERE THE MODIFICATION LIVES IN THE INERTIAL SECTOR — THE RESPONSE OF TEST BODIES — NOT IN A SINGLE SPACETIME METRIC. LIGHT DOES NOT AUTOMATICALLY FEEL THE SAME EFFECTIVE GEOMETRY AS SLOW MATTER. IN THIS PROGRAM THE PHOTON SECTOR IS A PUBLISHED **disformal** COMPLETION,

$$\tilde{g}_{\mu\nu} = g_{\mu\nu} + B u_\mu u_\nu,$$

WITH B FIXED BY THE SAME KERNEL $K(\square_u)$ THAT CONTROLS THE INERTIAL RESPONSE (THE LENSING NO-GO, DOI 10.5281/ZENODO.21418816, AND THE DISFORMAL COMPLETION; THE MI FIELD-THEORY RESULTS, DOI 10.5281/ZENODO.21403470). CONSEQUENTLY, **lensing- $a_0 = \text{dynamical-}a_0$** **is a prediction of that disformal construction, not an automatic consequence of the low-acceleration kernel**. THIS HAS TWO HONEST IMPLICATIONS, IN BOTH DIRECTIONS:

1. **Bonus test.** THE KiDS AGREEMENT IS A *CONSISTENCY TEST OF THE DISFORMAL PHOTON SECTOR*: IT IS A NON-TRIVIAL CHECK THAT LENSING AND DYNAMICS SHARE THE SAME SCALE, WHICH THE MODIFIED-INERTIA COMPLETION HAS TO ARRANGE AND COULD IN PRINCIPLE HAVE FAILED. THAT IS A GENUINE FEATHER IN THE COMPLETION’S CAP.

2. **Not an independent derivation.** PRECISELY BECAUSE THE EQUALITY IS ENGINEERED BY THE DISFORMAL SECTOR, THE KiDS LEG IS **not an independent derivation of “ Λ sets a_0 .”** IT IS CONSISTENT WITH THAT STATEMENT GIVEN THE DISFORMAL COMPLETION; IT DOES NOT ESTABLISH IT FROM LENSING ALONE. THE GEOMETRY LEG IS THEREFORE “INDEPENDENT” IN ITS SYSTEMATICS (SHEAR/PHOTO- z VS. KINEMATICS) BUT *CONDITIONAL* IN ITS *THEORY* (IT PRESUMES THE DISFORMAL SECTOR).

STATING THIS PLAINLY IS THE POINT: THE TRIANGLE IS A STRONG CONSISTENCY WEB, AND THE VALUE OF a_0 TOGETHER WITH Z REMAIN **posited**, NOT DERIVED. THE THREE LEGS TEST THE *UNIVERSALITY OF THE SCALE AND ITS INVERSION TO Λ* , NOT THE ORIGIN OF THE NUMBER.

7. Discussion

What the triangle buys. THREE CHANNELS — KINEMATICS, LENSING GEOMETRY, EXPANSION — WITH ORTHOGONAL SYSTEMATIC BUDGETS AGREE ON a_0 TO \$22–30%, AND ALL INVERT TO Λ WITHIN A FACTOR \$1.6 OF THE PLANCK VALUE, WITH PLANCK SITTING 0.91σ FROM THE LENSING CENTRAL. NOTABLY, THE DYNAMICS AND GEOMETRY LEGS LAND ON THE SAME *HIGH SIDE* OF THE BOX (1.18 AND 1.20), WHILE THE EXPANSION LEG AND THE CANONICAL FOOTING LAND LOW (0.92–0.99 AND 0.94). THIS IS A SELF-CONSISTENT PICTURE: A MODEST \$20% UPWARD SHIFT IN THE MEASURED GALACTIC/LENSING a_0 RELATIVE TO THE PURE- Λ CANONICAL VALUE MAPS, THROUGH $\Lambda \propto a_0^2$, TO THE FACTOR-\$1.6 EXCESS IN THE INFERRED Λ — THE SAME “HIGH-SIDE” BEHAVIOR IN BOTH THE DYNAMICS AND THE GEOMETRY LEGS, AND COMFORTABLY WITHIN THEIR SYSTEMATICS.

A link Λ CDM does not supply. IN Λ CDM THE GALACTIC a_0 EMERGES FROM BARYONIC-FEEDBACK-REGULATED HALO STRUCTURE AND HAS NO BUILT-IN NUMERICAL TIE TO THE FRIEDMANN Λ ; THE COINCIDENCE $a_0 \sim c\sqrt{\Lambda}$ IS, WITHIN THAT MODEL, A COINCIDENCE. THE FRAMEWORK HERE MAKES THE TIE STRUCTURAL VIA $\Lambda = 32\pi a_0^2/c^4$, AND THE TRIANGLE IS THE OBSERVATIONAL FACE OF THAT TIE. WE CLAIM THIS AS A *MOTIVATING CONSISTENCY*, NOT AS EVIDENCE THAT THE FRAMEWORK IS CORRECT — THE VALUE OF a_0 AND Z ARE POSITED.

What sharpens it. IN ORDER OF LEVERAGE: (1) THE TABULATED KiDS PER-POINT DATA, ENABLING A CLEAN LATE-TYPE FREE-FIT — A MODEST UPGRADE TO \$45% GIVEN THE SCATTER AMPLIFICATION, SECTION 4, CHIEFLY VALUABLE FOR REMOVING READING ERROR AND ENABLING THE LATE-TYPE CUT; (2) REDUCTION OF THE SPARC BOX BELOW \$20% VIA BETTER Υ_* AND INCLINATION CONTROL, WHICH THE PAIRWISE a_0 -LINE ESTIMATOR IS DESIGNED TO EXPLOIT; (3) TIGHTENING THE SNe/EXPANSION LEG THROUGH THE H_0

RANGE, WHICH CURRENTLY DRIVES MOST OF THE 0.92–0.99 SPREAD; AND (4) AN INTERNAL TEST OF THE DISFORMAL SECTOR THAT WOULD CONVERT THE GEOMETRY LEG FROM “CONDITIONAL ON THE COMPLETION” TOWARD “INDEPENDENT.” ANY ONE OF THESE NARROWS THE TRIANGLE. WE FLAG HONESTLY THAT THE REACH TOWARD A $\sim 10\%$ CROSS-CHECK WOULD BE CARRIED ALMOST ENTIRELY BY THE **SPARC-box (2) and SNe/expansion (3) legs**: WITH THE LENSING FREE-FIT NOW CAPPED AT $\sim 45\%$ FOR THE CLEAN LATE-TYPE SAMPLE (SECTION 4), THE GEOMETRY LEG NO LONGER CONTRIBUTES TO REACHING $\sim 10\%$ IS ASPIRATIONAL FOR THE DYNAMICS+EXPANSION PAIR, WITH LENSING SUPPLYING INDEPENDENCE RATHER THAN TIGHTNESS.

Caveats restated. THE KIDS LEG IS A *CONSISTENCY*, NOT A FREE-FIT (BROUWER ADOPTED a_0); A FIGURE FREE-FIT IS READING/SCATTER-LIMITED TO A FACTOR ~ 2 ; THE $\geq 6\sigma$ EARLY/LATE SPLIT STRAINS A SINGLE UNIVERSAL a_0 AND CONFINES THE CLEAN CASE TO LATE-TYPE DISKS; AND THE DISFORMAL WRINKLE CAPS THE GEOMETRY LEG’S INDEPENDENCE. NONE OF THESE IS FATAL, AND NONE IS HIDDEN.

8. Conclusion

THREE OBSERVATIONAL ROADS TO THE LOW-ACCELERATION SCALE — ROTATION-CURVE **dynamics**, WEAK-LENSING **geometry**, AND THE **expansion** HISTORY — MEET AT $a_0 \approx (0.92\text{--}1.20) \times 10^{-10} \text{ m s}^{-2}$, A $\sim 22\text{--}30\%$ SPREAD THAT LIES ENTIRELY INSIDE THE SPARC SYSTEMATICS BOX AND INVERTS, THROUGH $\Lambda = 32\pi a_0^2/c^4$, TO A COSMOLOGICAL CONSTANT WITHIN A FACTOR ~ 1.6 of the Planck value (— PLANCK) AT 0.91σ FROM THE LENSING CENTRAL). BECAUSE THE THREE LEGS’ SYSTEMATICS ARE MUTUALLY ORTHOGONAL, THEIR AGREEMENT IS A NON-TRIVIAL CONSISTENCY WEB ON THE STATEMENT THAT THE ACCELERATION SCALE IS UNIVERSAL AND SET BY $\sqrt{\Lambda}$.

WE HAVE BEEN DELIBERATE ABOUT SCOPE. THIS IS A **consistency + method + forecast note, not a free-fit measurement**. THE LENSING LEG IS A *CONSISTENCY* — BROUWER ADOPTED a_0 AND TESTED AGREEMENT — AND A GENUINE LENSING FREE-FIT AWAITS PER-POINT DATA (FORECAST: $\sim 45\%$ FOR THE CLEAN LATE-TYPE SAMPLE, SINCE THE $a_0 \propto g_{\text{obs}}^2$ SCATTER AMPLIFICATION LEAVES STATISTICAL AND SYSTEMATIC ERRORS COMPARABLE). THE $\geq 6\sigma$ EARLY/LATE-TYPE SPLIT IS A REAL STRAIN THAT MAKES LATE-TYPE DISKS THE CLEAN CASE. AND IN MODIFIED INERTIA THE LENSING–DYNAMICS EQUALITY IS A *PREDICTION OF THE DISFORMAL PHOTON SECTOR*, NOT AUTOMATIC, SO THE GEOMETRY LEG IS INDEPENDENT IN ITS SYSTEMATICS BUT CONDITIONAL IN ITS THEORY. THE VALUE OF a_0 AND THE CONSTANT $Z = \sqrt{32\pi/3}$ ARE **posited, not derived**; THE TRIANGLE TESTS UNIVERSALITY AND SCALE, NOT THE ORIGIN OF THE NUMBER. WHAT THE NOTE BOOKS IS THE METHOD AND THE CURRENT STATE OF THE THREE-CHANNEL AGREE-

MENT, SO THAT THE SHARPENING STEPS — CHIEFLY THE TABULATED KIDS
FREE-FIT — HAVE A CLEAR TARGET.

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REPRODUCIBILITY: EVERY DIMENSIONAL QUANTITY IN THIS NOTE IS EMITTED BY THE COMMITTED SCRIPT `KIDS_RAR_LAMBDA.PY` (EXIT 0). a_0 IS CARRIED ON BOTH FOOTINGS (CANONICAL $a_0 = cH_\Lambda/Z = 9.355 \times 10^{-11}$; ALTERNATIVE $a_0 = 1.1305 \times 10^{-10}$). NO CLAIM OF DERIVATION IS MADE FOR a_0 OR Z .